

# Yutian Zhang

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## EDUCATION

### The Hong Kong University of Science and Technology

2024 - Present

*Ph.D. in Electronic and Computer Engineering | Advisor: Prof. Mansun Chan*

- Research focus: device-circuit DTCO of CFET SRAM under practical layout, stability, and parasitic constraints.

### The Hong Kong University of Science and Technology

2020 - 2024

*B.Eng. in Electronic and Computer Engineering*

- GPA: 3.62/4.0; Major GPA: 3.91/4.0; Rank: 3/133 (top 2%).

## PUBLICATIONS

[1] Y. Zhang, K. Sarfraz, and M. Chan, "DTCO-based Hybrid Rail 8T Complementary FET SRAM Design towards Advanced Node," Accepted.

[2] Y. Zhang, K. Sarfraz, and M. Chan, "DTCO-based Monolithic Vertical Four-layer 6T Complementary FET SRAM Design," Accepted.

[3] Y. Zhang, J. Fang, K. Sarfraz, and M. Chan, "CFET SRAM Design Manuscript," Manuscript under review.

## RESEARCH EXPERIENCE

### Device-Circuit DTCO of Complementary FET SRAM

2024 - Present

*Ph.D. Researcher, HKUST | Advisor: Prof. Mansun Chan*

- Study CFET as a circuit-technology design space where device architecture, layout, tier configuration, parasitics, and circuit metrics are co-optimized.
- Develop SRAM-centered DTCO studies connecting vertical CFET integration to density, access performance, stability, energy, and array-level PPA outcomes.
- Investigate CFET SRAM layout and sizing options, including cell organization and read/write stability tradeoffs, with detailed concepts disclosed after publication.

### SRAM and DRAM Based In-Memory Computing

Feb 2023 - 2024

*Undergraduate Research Assistant, HKUST | Advisor: Prof. Shao Qiming*

- Designed 6T SRAM for binary neural-network operation using split word lines, reducing average word-line current by approximately 30%.
- Designed a small-offset voltage-mode sense amplifier and evaluated write-noise-margin improvement through Monte Carlo analysis.

### Analog Circuit Design Internship

Jul 2023 - Aug 2023

*InstonTech, Suzhou, China*

- Analyzed conventional 6T SRAM behavior and proposed a 9T SRAM design with improved static-noise-margin characteristics.
- Reduced write power through an improved latch-based precharge circuit and supported module-level reverse engineering from netlist information.

## SELECTED PROJECTS

### Hybrid Rail 8T CFET SRAM; Monolithic Four-layer 6T CFET SRAM; Ongoing CFET SRAM

#### DTCO Study

- Accepted and under-review CFET SRAM works covering hybrid rail design, monolithic vertical four-layer SRAM, and broader device-circuit DTCO questions.
- Future Ph.D. modules will be disclosed after the corresponding work is published.

## TECHNICAL SKILLS

|                  |                                                                 |
|------------------|-----------------------------------------------------------------|
| Device / circuit | CFET SRAM design, DTCO, SRAM stability analysis, PPA evaluation |
| EDA / simulation | Cadence Virtuoso, HSPICE/Spectre, Silvaco, VHDL                 |
| Programming      | Python, MATLAB, C/C++                                           |

## HONORS

### Dean's List

*The Hong Kong University of Science and Technology*